

Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization

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eLife Assessment

This study presents a meta-analysis of two independent genome-wide association studies (GWAS) that investigate the role of plasma proteins as potential biomarkers for enhancing the early detection of prostate cancer (PCa). The results provide **useful** confirmatory data that support existing evidence currently published. The evidence is **incomplete**: the study does not provide a comprehensive synthesis of all currently published work, does not explore other clinical outcomes related to prostatic disease, and its findings have not been validated through an external cohort study. These shortcomings notwithstanding, the work may be of interest to researchers studying correlates and predictors of prostate cancer risk.

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Abstract

Background

Due to the limitations in specificity of current diagnostic methods for prostate cancer (PCa), more reliable biomarkers are needed to explore for improving early detection. Plasma proteins represent a promising source of biomarkers, therefore understanding the causal relationships between specific plasma proteins and PCa could be conducive to identify novel biomarkers and therapeutic targets for PCa prevention and treatment.

Methods

We performed a meta-analysis of two independent genome-wide association studies (GWASs) including 94,397 individuals with PCa and 192,372 controls. A mendelian randomization (MR) supplemented by colocalization analysis was conducted, using *cis*-acting variants on 4,907 plasma proteins from deCODE Genetics (N=35,559) and 2,940 plasma proteins from UK

Biobank Pharma Proteomics Project (UKB-PPP) (N=54,219). Then, the biological pathway analysis and druggability evaluation of the risk proteins were further performed.

Results

Five possible susceptibility loci (JAZF1, PDILM5, WDPCP, EEFSEC, and TNS3) for PCa were identified through the meta-analysis of GWASs. Among 3,722 plasma proteins, 193 proteins were associated with PCa risk, of which 20 high-risk proteins, including KLK3, were validated in both the deCODE and UKB-PPP cohorts. Functional annotation of these genes encoding proteins confirmed enrichment of immune response, inflammatory response, cell-cell interaction and so on. Genetic colocalization and druggable genome analyses also identified several potential drug targets for PCa, such as HSPB1, RRM2B and PSCA.

Conclusions

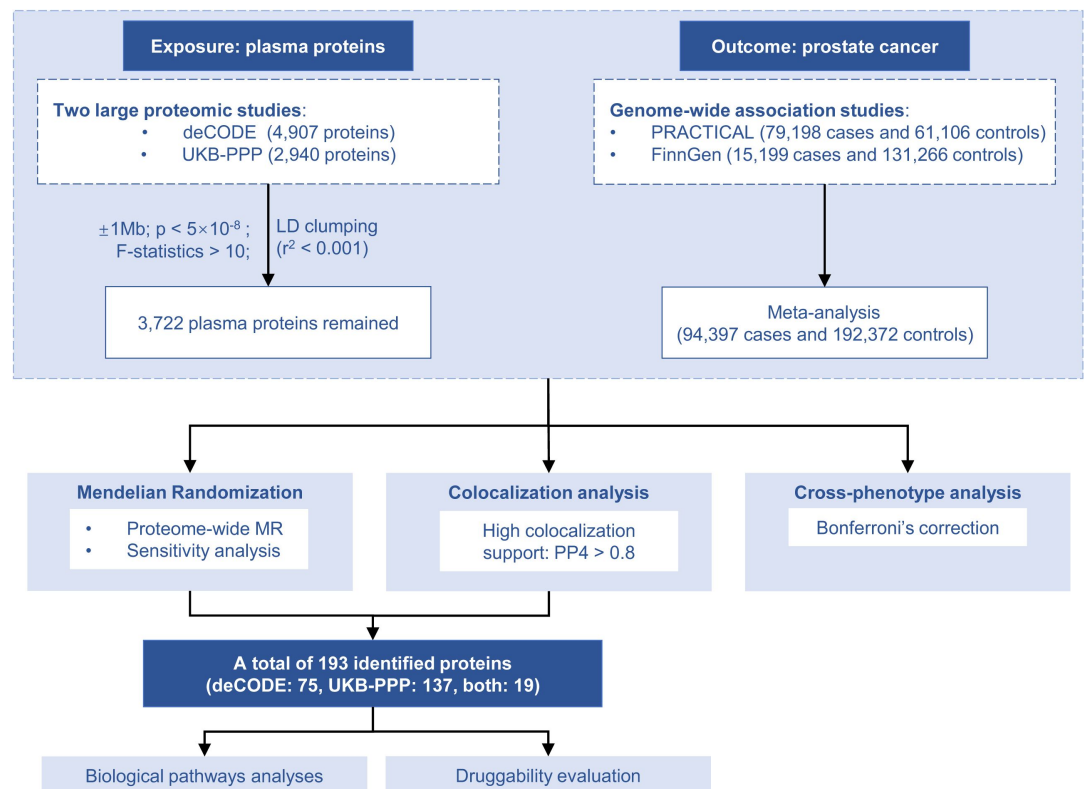
We identified novel variants as well as several protein biomarkers linked to PCa risk and indicated pathways associated with PCa, which offered new insights into PCa etiology and contributed to development of novel biomarkers for early detection and potential therapeutic interventions.

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Graphical abstract

A comprehensive study design summarized as follows.



Introduction

Prostate cancer (PCa) is the second most common malignancy with an estimation of 1.5 million new cases in 2022, which is account for 14% of total cancer diagnosed in men worldwide (Ferlay J, 2024). Even more worrying is that the number of new cases of PCa are projected to 2.9 million by 2040 (James et al., 2024). Despite advancements in medical technology, early detection of PCa remains a significant challenge due to the limitations in specificity of current diagnostic methods, such as prostate-specific antigen (PSA) testing (Kalavacherla et al., 2023; Tikkinen et al., 2018). Therefore, there is an urgent need for more accurate and reliable biomarkers that can improve early detection and prognostic evaluations of PCa.

Plasma proteins have key roles in the development and progression of PCa, such as interleukin (IL)-6 (Deichaite et al., 2022), insulin-like growth factor (IGF)-1 and IGF-binding protein (IGFBP)-1 (Cao et al., 2015). However, observational studies exploring the association between plasma proteins and PCa risk are often limited by confounding factors and selection bias, making it challenging to establish a clear causal relationship.

To overcome these limitations, Mendelian randomization (MR) offers a robust methodological approach. This approach utilizes the principle of random assortment of genes from parents to offspring, using genetic variation as an instrumental variable, which mimics the randomization process in a controlled trial, thereby minimizing the impact of reverse causality (Davies, Holmes, & Smith, 2018). With the recent development of proteomics technology, several large-scale proteomic studies have identified over 18,000 protein quantitative trait loci (pQTLs) covering more than 4800 proteins (Feringstad et al., 2021; Pietzner et al., 2021). By using MR, researchers can gain a deeper comprehension of whether particular plasma proteins hold a causal nexus with PCa, thereby potentially unraveling novel biomarkers and targets for the prevention and treatment of PCa.

In this study, we performed a meta-analysis of two genome-wide association studies (GWAS) on PCa (PRACTICAL and FinnGen) for a total sample size of 94,397 cases and 192,372 controls. Based on GWAS summary statistics data from deCODE and UKB-PPP cohorts, we further performed a protein-wide MR (PW-MR) study, supplemented by colocalization analysis, to explore the casual relationship between plasma proteins and PCa risk. Moreover, we indicated biological processes and pathways associated with PCa, and evaluated the druggability of risk proteins. We aimed to identify novel plasma proteins biomarkers for PCa, which could address the limitations of current diagnostic methods and offer new insights into the biological mechanisms of PCa and potential therapeutic targets for intervention.

Methods

Data sources for plasma proteins

We selected *cis*-SNPs associated with plasma proteins as instrumental variables from two large-scale GWASs in the deCODE Genetics (Ferkingsstad et al., 2021 [↗](#)) and UKB-PPP (Sun et al., 2023 [↗](#)). *Cis*-SNPs were defined as SNPs within a vicinity of ± 1 Mb around the gene encoding the protein. From these SNPs, only those with a minor allele frequency of $\geq 1\%$ that were genome-wide significance ($p < 5 \times 10^{-8}$) and considered independent (linkage disequilibrium $r^2 < 0.1$ in 1000G) were retained. deCODE Genetics conducted proteomic profiling on blood plasma samples from 35,559 Icelanders using the SomaScan platform and collected data on 4907 aptamers (Ferkingsstad et al., 2021 [↗](#)). For the two-sample MR analysis, we selected *cis*-SNPs as instrumental variables for 1778 proteins. Likewise, *cis*-SNPs for 1944 plasma proteins were obtained from the UKB-PPP where 2940 proteins were measured among 54,219 Europeans using the Olink platform (Sun et al., 2023 [↗](#)). The proteins with positive MR results were included in the colocalization analysis.

Data sources for Prostate Cancer

GWAS summary statistics for PCa were obtained from the PRACTICAL consortium (Schumacher et al., 2018 [↗](#)) and FinnGen study. The PRACTICAL (Prostate Cancer Association Group to Investigate Cancer Associated Alterations in the Genome) consortium included 79,198 cases and 61,106 controls. We used the data on PCa from the FinnGen study R10 in this analysis, which comprised 15,199 cases and 131,266 controls. All participants of these two cohorts were of European ancestry. In the MR analysis, we treated the PRACTICAL consortium as the discovery study and the FinnGen R10 study as the replication. To increase the effectiveness, we performed a fixed-effect GWAS meta-analysis of the two GWASs using the METAL package. The quantile-quantile plot was generated using the “qqman” package in R software (4.3.3). The R package “gassocplot” was used to plot regional association plots for the top SNP at each of the genome-wide significant loci identified.

Cross-phenotype analysis

Through the interactive Cross-Phenotype Analysis of GWAS database (iCPAGdb) (Wang et al., 2020 [↗](#)), a new platform for cross-phenotype analysis, we explored the genetic correlation between PCa and other traits. We analyzed the genetic correlations between PCa and 3793 traits using data from the National Human Genome Research Institute-European Bioinformatics Institute (NHGRI-EBI) GWAS catalog (Wang et al., 2020 [↗](#)). Through the use of ancestry LD-specific association data, iCPAGdb performs cross-phenotype enrichment analyses. iCPAGdb reveals signals of pairwise traits and shared signals by analyzing traits associations with LD proxy SNPs. The output data will show results from Fisher’s exact test with adjustment for 5% false discovery rate (FDR) and also Bonferroni’s, Jaccard’s, Sorensen’s, and Chao-Sorensen similarity indexes.

MR analysis

We conducted a MR analysis with plasma proteins as the exposure variable and PC as the outcome variable. We utilized the R package “TwoSampleMR” in R software (4.3.3) for the MR analysis. When only one SNP was available for a particular protein, we applied the Wald ratio test. Inverse-variance weighted (IVW) was used as the main analysis method for two or more SNPs available. Considering multiple testing, we employed the 5% False Discovery Rate (FDR) method for P-value correction. $P < 0.05$ was considered statistically significant. MR-Egger and weighted median methods were used as supplementary analysis methods. Cochran’s Q statistic test using the MR-Egger method was used to determine the heterogeneity between the genetic variants. We also performed MR-Egger regression intercept to detect and adjust the directional horizontal pleiotropy (Hemani, Bowden, & Smith, 2018 [↗](#)), $P < 0.05$ was considered as the presence of directional pleiotropy and thus removed from the further analyses.

Colocalization analysis

The “coloc” package was used to perform Bayesian colocalization analysis to investigate if the associations between plasma proteins and PCa were driven by linkage disequilibrium (LD) (Giambartolomei et al., 2014 [↗](#)). For proteins with positive MR results, the Bayesian method assessed the support for the following five exclusive hypotheses: 1) no association with either trait; 2) association with trait 1 only; 3) association with trait 2 only; 4) both traits are associated, but distinct causal variants were for two traits; and 5) both traits are associated, and the same shares causal variant for both traits (Foley et al., 2021 [↗](#)). The posterior probability is provided for each hypothesis (H0, H1, H2, H3, and H4). In this analysis, we set prior probabilities of the SNP being associated with trait 1 only (p1) at 1×10^{-4} ; the probability of the SNP being associated with trait 2 only (p2) at 1×10^{-4} ; and the probability of the SNP being associated with both traits (p12) at 1×10^{-5} . Two signals were considered to have strong evidence of colocalization if the posterior probability for shared causal variants (PH4) was ≥ 0.8 . The analysis was performed in R software (4.3.3).

Biological pathway analysis

The GENE2FUNC tool in FUMA (Watanabe, Taskesen, van Bochoven, & Posthuma, 2017 [↗](#)) web application programming interface version 1.5.2 were used to investigate functional annotation and enrichment analysis of the genes coding for the 193 unique proteins identified by the PW-MR approach. This analysis involved calculating the log2 fold-change expression for each gene across 54 tissue types from the GTEx (Genotype-Tissue Expression) database. We conducted gene enrichment analysis to identify overrepresented biological processes using the GO (Gene Ontology) database, which helped determine the biological processes most relevant to the genes identified. To investigate the involvement of the identified genes in functional and signaling pathways, we used the KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway database. Enrichment results were considered significant if they passed 5% FDR P-value threshold. Additionally, only enrichments involving at least two overlapping genes with the gene sets were considered to ensure robustness and biological relevance.

Druggability evaluation

The analysis of drug targets for the significant proteins identified in the MR Analysis was conducted using data from OpenTargets (Ochoa et al., 2023 [↗](#)) (v.22.11), which is publicly accessible. We selected all drugs for which there was evidence for an association with the protein of interest.

Results

Meta-analysis of the Genome-Wide Association Studies for Prostate Cancer

We conducted a meta-analysis combining two GWAS studies with a collective sample size of 94,397 individuals with PCa and 192,372 controls, aiming to identify genetic variants linked to PCa. The associations and assessment of SNPs heterogeneity that passed the genome-wide P-value threshold at these loci with PCa were presented in Supplementary Table 1. We found 5 genetic risk loci contained at least one SNP passing the genome-wide significance threshold of $P \leq 5 \times 10^{-8}$: JAZF1, PDILM5, WDPCP, EEFSEC, and TNS3 (**Figure 1**). Among them, PDILM5, WDPCP, EEFSEC and TNS3 were promising candidates as novel susceptibility loci associated with PCa. Supplementary Figure 1 displayed the associated quantile-quantile plot. The LocusZoom plots of the top SNPs at JAZF1, PDILM5, WDPCP, EEFSEC and TNS3, along with their genomic location, GWAS P values and recombination rate with neighboring SNPs were visualized in Supplementary Figure 2. In summary, this GWAS meta-analysis discovered genetic variations in one recognized PC-associated loci and four potential novel loci, providing a reliable dataset for MR analyses.

Cross-phenotype Analysis of Prostate Cancer

The cross-phenotype analysis was designed to systematically identify phenotypic traits that share genetic or molecular pathways with PCa, thereby uncovering pleiotropic mechanisms or shared risk factors. We used the iCPAGdb to conduct this analysis with PCa genome-wide significant SNPs ($P \leq 5 \times 10^{-8}$). The iCPAGdb offered an improved algorithm for identifying cross-phenotype associations by using pre-computed ancestry-specific LD databases and integrating genetic data from 3793 traits in the NHGRI-EBI GWAS catalog(Wang et al., 2020). Cross-phenotype analysis of PCa-associated SNPs identified 117 traits significantly linked to PCa after Bonferroni's correction(Supplementary Table 2). As shown in **Figure 2**, the strong association with PSA measurement ($P=7.16 \times 10^{-43}$) validated our genetic instruments, while the enrichment of metabolic traits such as BMI-adjusted waist-hip ratio (WHR) ($P=1.32 \times 10^{-26}$), high density lipoprotein (HDL) cholesterol measurement ($P=2.38 \times 10^{-22}$) and body mass index ($P=1.71 \times 10^{-20}$) aligned with epidemiological evidence linking obesity to aggressive PCa. Notably, the strong association with balding ($P=3.01 \times 10^{-22}$) reinforced the androgen axis in PCa pathogenesis and the correlation with heel bone mineral density ($P=2.02 \times 10^{-19}$) was consistent with the clinical observations of osteoblastic bone metastasis in advanced PCa. These cross-phenotype associations collectively highlight PCa as a systemic disorder involving androgen signaling, metabolic dysregulation and skeletal interactions, providing novel insights into its multifactorial etiology.

Proteome-Wide Mendelian Randomization Studies of Prostate Cancer

The genetic association summary statistics of 35,559 Icelanders from deCODE Genetics and 54,219 Europeans from the UKB-PPP were utilized to investigate the relationship between PCa and plasma proteins. Our genetic instrument selection strategy enabled us to examine 1778 and 1944 proteins from deCODE and UKB-PPP (Supplementary Table 3A and B). Using the Wald ratio or IVW method, a total of 193 unique plasma proteins were significantly associated with PCa after multiple tests with a 5% FDR correction. This analysis yielded 137 proteins in UKB-PPP more than 76 proteins in deCODE (**Figure 3A**). The results of Heterogeneity test based on Q statistics showed little evidence of heterogeneity. Furthermore, no significant intercept was detected, implying that there was no directional pleiotropy observed.

Figure 1.

Manhattan plot of PCa GWAS meta-analysis.

The genetic regions containing top SNPs related to PCa are depicted. The red dashed line signifies the genome-wide significance threshold of 5.0×10^{-8} .

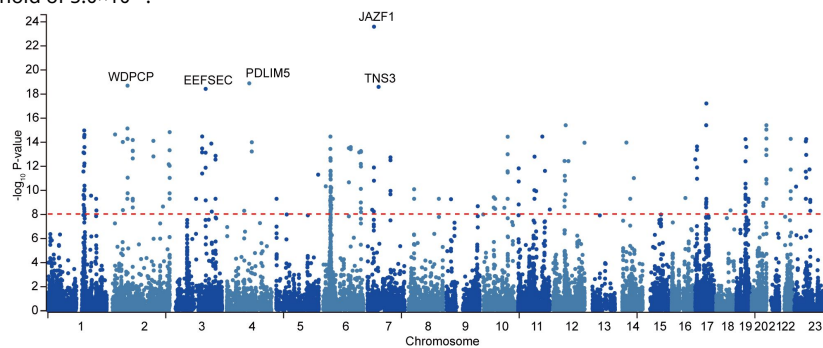
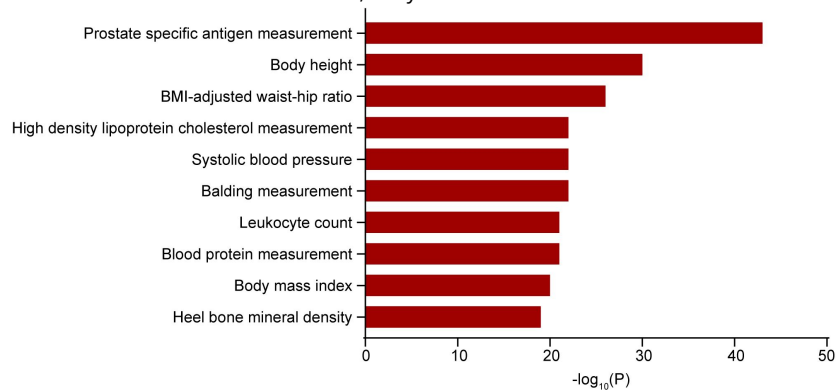


Figure 2.

The top 10 significant cross-phenotype associations with PCa at a 5% FDR.

The x-axis represents the P value of the correlation. BMI, body mass index.



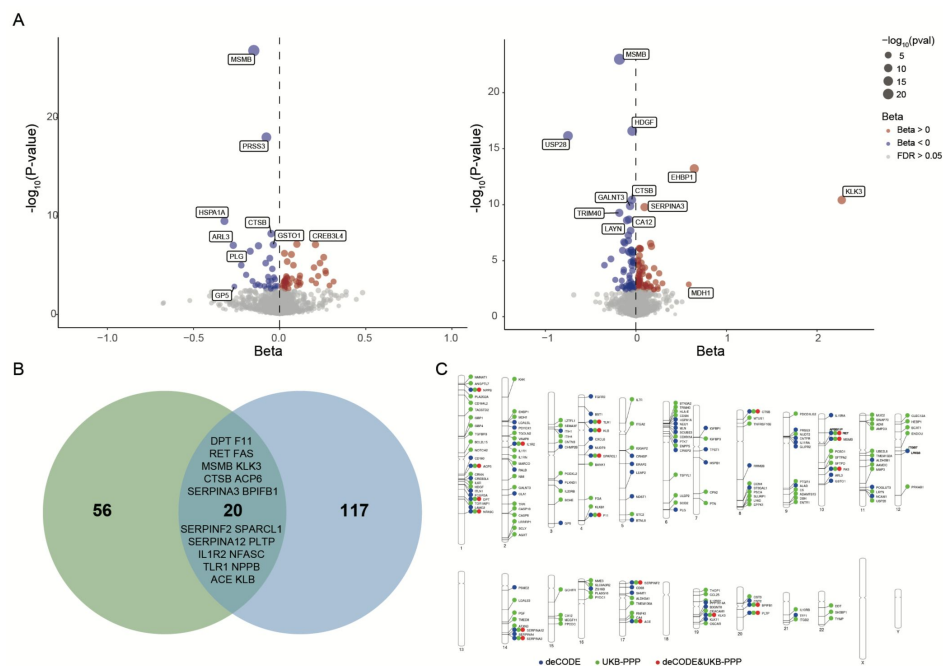


Figure 3.

Result of PW-MR on the associations between plasma proteins and the risk of PCa.

(A) Volcano plot of PCa PW-MR study using deCODE (the left side) and UKB-PPP (the right side) cohorts. Annotated proteins passed the 5% FDR IVW P-value threshold. The blue and red colors represent a negative and positive effect, respectively. (B) Venn diagram depicting proteins associated with PCa in deCODE and UKB-PPP. (C) PhenoGram of PCa PW-MR study significant associations. The blue dots and the green dots represent the deCODE and UKB-PPP specific proteins, respectively. The red dot represents both simultaneously.

After FDR correction, 20 proteins were detected in both data sets and the identified associations were consistent (**Figure 3B**). Genetic prediction indicated that 11 of these 20 proteins were positively associated with the risk of PCa as well as remaining 9 proteins were negatively associated, suggesting that these 9 proteins maybe protective factors against PCa (Supplementary Table 4). A PhenoGram depicted the chromosomal location of the 193 unique identified proteins in deCODE and UKB-PPP studies (**Figure 3C**).

Colocalization analysis

We performed colocalization analyses of proteins significantly expressed in deCODE and UKB-PPP studies with PCa. It was observed that 4 proteins in deCODE study and 7 proteins in UKB-PPP were colocalized with PCa associations with high support of evidence ($PPH4 \geq 0.8$) (**Table 1**), suggesting that these 10 plasma proteins might serve as potential targets for treating PCa. Among them, SERPINA3 showed strong colocalization evidence in both deCODE ($PPH4=0.952$) and UKB-PPP ($PPH4=0.951$) studies. This analysis identified 1 causal variant in deCODE (rs61976125) and 1 causal variant in Fenland (rs6575449) (**Figure 4**).

Gene-Based Association and Pathway Analyses

We used the GENE2FUNC tool available in FUMA to explore the biological significance, functional implications and tissue-specific expression of the genes identified from our GWAS. These genes appeared to be significantly enriched in inflammatory and immune pathways (such as defense, immune and inflammatory response), as well as cell interaction and signaling pathways (such as interaction between organism and cell adhesion) (**Figure 5A**). The KEGG pathway enrichment analysis revealed several pathways that were significantly enriched, such as cytokine-cytokine receptor interaction, p53 signaling pathway, JAK-STAT signaling pathway, pathways in cancer and apoptosis (**Figure 5B**). These analyses indicated potential biological processes and mechanisms associated with PCa. The 193 unique genes were predominantly expressed in the lymphocytes, blood, liver and prostate (Supplementary Figure 3).

Druggability of identified proteins

Exploring new therapeutic opportunities for PCa based on genetic information is crucial for developing targeted treatments. In our study, we examined the druggability of genes and proteins identified in MR analyses using OpenTargets databases. We investigated 128 unique drugs targeting 45 identified proteins (Supplementary Table 5). Among these, three drugs (APATORSEN, TRIAPINE and MK-4721) currently in clinical trials were intended for PCa treatment, with each targeting HSPB1, RRM2B and PSCA, respectively. Notably, the effects of these drugs on their respective protein targets align with the directions indicated by our MR results, suggesting a consistency between genetic evidence and therapeutic potential. In addition, several other identified targets, such as RET, FGFR3, NCAM1, TYMP, TNFRSF10B, MMP3, TACSTD2 and NOTCH2, were implicated in various cancers and present potential therapeutic avenues.

Discussion

Genetic Susceptibility Loci and Functional Mechanisms

In this study, we conducted a comprehensive meta-analysis of two GWAS for PCa, identifying significant genetic loci associated with PCa risk. The combined sample size of 94,397 cases and 192,372 controls revealed one known (JAZF1) and four potential novel (PDLIM5, WPCP, EEFSEC and TNS3) susceptibility loci of PCa. These genetic findings not only expand our understanding of PCa heritability but also bridge the gap between genetic risk and downstream molecular mechanisms.

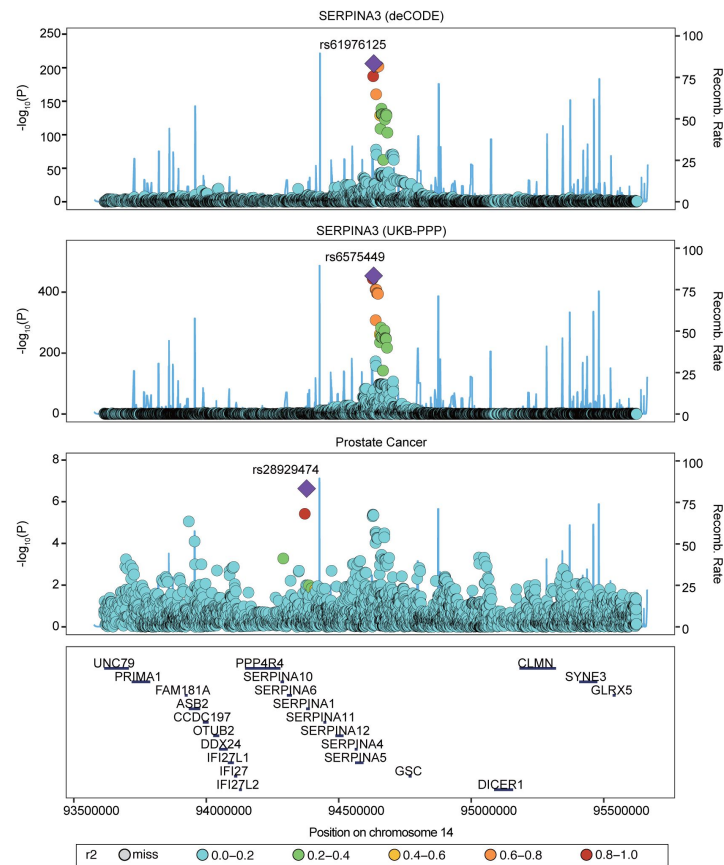


Figure 4.

Colocalization plot of SERPINA3 variants associated with PCa in deCODE and UKB-PPP.

Variants are color-coded based on their linkage disequilibrium (LD) with the lead SNP (the variant with the lowest p-value). The lead SNP is highlighted in red. Other variants are colored from blue to yellow, indicating decreasing LD with the lead SNP.

GWAS	Outcomes	Proteins	Mendelian randomization			Colocalization
			OR (95%CI)	P value	P value after FDR adjustment	Analysis PH4
deCODE	Prostate Cancer	MSMB	0.86(0.84,0.88)	1.56E-27	2.82E-24	0.999
		POGLUT3	0.89(0.83,0.96)	1.16E-03	2.44E-02	0.998
		PRSS3	0.93(0.91,0.94)	1.00E-18	6.03E-16	0.956
		SERPINA3	1.11(1.06,1.15)	7.18E-08	1.05E-05	0.952
UKB-PPP	Prostate Cancer	USP28	0.47(0.40,0.56)	7.27E-17	2.89E-14	0.999
		KLK3	9.70(4.94,19.03)	3.83E-11	6.91E-09	0.999
		IGFBP3	1.07(1.05,1.09)	2.04E-11	4.49E-09	0.997
		CASP10	1.22(1.11,1.34)	4.73E-05	1.54E-03	0.988
		HDGF	0.96(0.95,0.97)	2.61E-17	1.29E-14	0.966
		SERPINA3	1.10(1.07,1.13)	1.64E-10	2.32E-08	0.951
		C5	1.19(1.11,1.27)	5.32E-07	4.06E-05	0.883

OR, odds ratio; CI, confidence interval; FDR, false discovery rate.

Table 1.

Analysis of Mendelian randomization and colocalization of significant proteins with PCa.

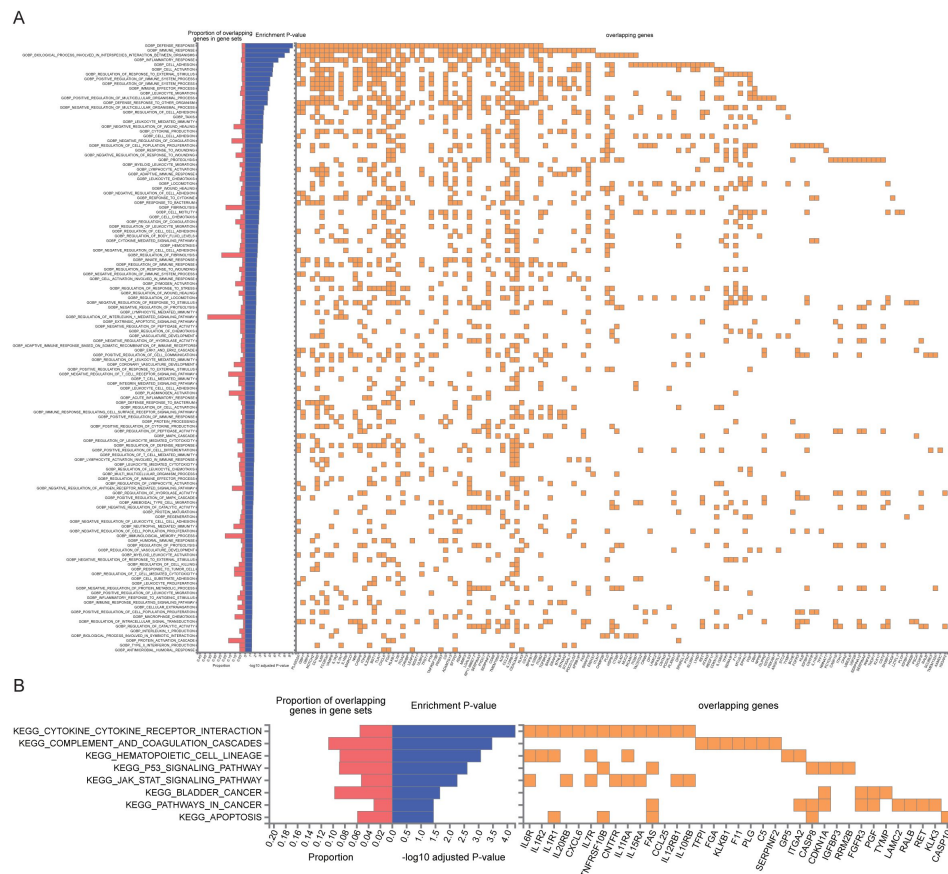


Figure 5.

Functional annotation of the genetic architecture of PCa.

(A) Biological processes and (B) KEGG pathway analysis of the 193 unique proteins identified in deCODE and UKB-PPP.

JAZF1, a transcriptional repressor linked to metabolic regulation (Rosario et al., 2023) and cellular proliferation (Sung et al., 2018), has been implicated in both type 2 diabetes and PCa risk through GWAS studies (Machiela et al., 2012; Sánchez-Maldonado et al., 2022). Its dual role highlights the interaction between metabolic dysregulation and oncogenesis, indicating that interventions targeting metabolic pathways may hold therapeutic promise for PCa.

PDLIM5 (PDZ and LIM domain 5), a cytoskeleton-associated protein, regulates cell migration and tumor progression by binding a variety of proteins through its specific domains. In PCa, PDLIM5 promotes epithelial-mesenchymal transition and migration of prostate cancer cells (Liu et al., 2017), while downregulating the expression of PDLIM5 may ultimately impede the progression of PCa (Xie et al., 2020), indicating its potential value in predicting the risk of advanced PCa. WPCP, a key effector of planar cell polarity (PCP) signaling, regulates actin cytoskeleton to maintain tissue architecture (Cui et al., 2013). Disruption of PCP signaling can cause abnormal cell behavior and tissue structure, leading to the development of cancer (Humphries & Mlodzik, 2018). EEFSEC is a member of the eukaryotic family of elongation factors and its role in tumor cells has been rarely reported. A functional study demonstrated that EEFSEC was significantly upregulated in PCa cells and high expression of EEFSEC was associated with poor prognosis in patients with PCa (Xu et al., 2021). TNS3 promotes cell migration and invasion through focal adhesion dynamics (Chen et al., 2017; Martuszevska et al., 2009; Zheng et al., 2021; Zuidema et al., 2022), which is critical in cancer metastasis. While these loci implicate diverse pathways, their collective impact may converge on tumor microenvironment (TME) reprogramming. To certain whether these 4 genes causative factors for PCa, genetic association studies with a larger sample size will be needed.

Multifactorial Drivers of Prostate Cancer

PCa arises from a complex interplay of genetic, metabolic and hormonal perturbations. In the cross-phenotype Analysis of PCa, we found multiple risk-associated traits including proteomic biomarkers (PSA), metabolic dysregulation (central obesity, HDL dysfunction, hypertension) and androgen signaling (balding), collectively underscoring PCa's multifactorial etiology.

PSA, a protein secreted by both normal and malignant prostate epithelial cells, has long been a cornerstone in the early detection and monitoring of PCa. However, the specificity and sensitivity of PSA as a diagnostic tool have been questioned due to its elevation in benign conditions such as prostatitis and benign prostatic hyperplasia (BPH), leading to unnecessary biopsies and substantial overdiagnosis (Gudmundsson et al., 2018; Ilic et al., 2018; Kalavacherla et al., 2023). Therefore, complementary biomarkers are urgently needed to improve diagnostic accuracy. Our PW-MR study revealed that genetically plasma MSMB levels were causally associated with PCa risk. Importantly, this aligns with clinical utility of urinary MSMB: a quadriplex urine panel (MSMB, TRPM8, AMACR, PCA3) improved diagnostic accuracy in PCa patients compared to PSA alone (Jamaspishvili et al., 2011). While plasma MSMB reflects systemic risk, urinary MSMB captures localized prostate pathology, together offering complementary insights into PCa biology. In conclusion, systematic integration of proteomic biomarkers identified through causal inference frameworks with PSA-based screening offers a path to resolve the persistent challenges of PCa diagnostic accuracy.

Emerging evidence implicates metabolic syndrome (MetS), which encompasses a cluster of conditions including central obesity, dyslipidemia and hypertension, may be a risk factor of PCa and may also worsen outcomes (Hernandez-Perez et al., 2022; Lifshitz, Ber, & Margel, 2021). The significant association between BMI-adjusted WHR and advanced PCa risk suggested that central obesity may play a critical role in the development of PCa (Genkinger et al., 2020; Perez-Cornago, Dunneram, Watts, Key, & Travis, 2022). Visceral adiposity may increase risk of advanced forms of PCa through changes in cytokines and growth factors, hormone regulation and metabolism (Doyle, Donohoe, Lysaght, & Reynolds, 2012; Himbert et al., 2017). These metabolic perturbations extend beyond obesity itself, intersecting with broader lipid and inflammatory

pathways that further modulate PCa risk. For instance, HDL cholesterol, while classically recognized for its cardioprotective roles in reverse cholesterol transport and antioxidant and anti-inflammatory activities (Rohatgi, Westerterp, von Eckardstein, Remaley, & Rye, 2021 [\[1\]](#)), may also antagonize prostate carcinogenesis. Epidemiological and preclinical studies found that HDL may suppress prostate carcinogenesis by reducing oxidative stress and the levels of pro-inflammatory molecules in cancer cells and TME (Ossoli, Wolska, Remaley, & Gomaschi, 2022 [\[2\]](#); Ruscica et al., 2018 [\[3\]](#)). However, in obesity, HDL's antioxidant and anti-inflammatory capacity is impaired due to altered composition and reduced functionality (Bacchetti, Morresi, Simonetti, & Ferretti, 2024 [\[4\]](#)), while adipose-derived ROS and cytokines further amplify oxidative damage (Balan, Halatiu, & Scridon, 2024 [\[5\]](#)). The loss of HDL protection and obesity-driven oxidative stress provides a physiological mechanistic basis for obesity-associated PCa aggressivity. This metabolic-inflammatory axis extends to hypertension, which is another MetS hallmark. Although results from previous studies of the association between hypertension and PCa development remain inconsistent (Christakoudi et al., 2020 [\[6\]](#); Liang et al., 2016 [\[7\]](#); Seretis et al., 2019 [\[8\]](#)), a MR analysis suggested that elevated systolic blood pressure might increase PCa risk through systemic inflammation (Stikbakke et al., 2022 [\[9\]](#)), a mechanism similar to that of central obesity and HDL dysfunction. Chronic inflammation contributes to a pro-tumorigenic environment by promoting cellular proliferation, DNA damage and resistance to apoptosis (Mantovani, Allavena, Sica, & Balkwill, 2008 [\[10\]](#)). Thus, MetS components likely converge on overlapping pathways to accelerate PCa progression, though the precise relationship between systolic blood pressure and PCa remains unclear. More research is needed to elucidate the exact biological pathways involved and to determine whether managing clinical features of MetS could serve as a preventive strategy for PCa.

The link between androgenic alopecia (AGA) and PCa risk further implicates androgen metabolism as a central driver, as both conditions are influenced by dihydrotestosterone (DHT) levels. In AGA, DHT binds to androgen receptors (AR) in the dermal papilla cells, causing follicular miniaturization and hair loss (Urysiak-Czubatka, Kmiec, & Broniarczyk-Dyla, 2014 [\[11\]](#)). Similarly, DHT-driven AR activation in prostate epithelial cells promotes proliferation and inhibits apoptosis, fostering tumorigenesis (Tong et al., 2022 [\[12\]](#)). However, the link between AGA and PCa may not be solely hormone-dependent. Genetic pleiotropy where variants in loci such as AR or SRD5A2 influence both balding and prostate carcinogenesis, which could partially explain this association (Hayes et al., 2005 [\[13\]](#); Hayes et al., 2007 [\[14\]](#)). Further studies are needed to confirm the causality.

Overall, our cross-phenotype analysis provided a comprehensive view of the diverse factors associated with PCa and enhanced our understanding of the disease's multifaceted nature. Future researches should focus on elucidating the underlying mechanisms of these associations and exploring their potential for integration into clinical practice.

Plasma Proteins Causal Links to Prostate Cancer

The PW-MR study supplemented by colocalization analysis identified 193 unique proteins causally associated with PCa, including 10 with robust colocalization evidence. These findings pinpoint both established and novel protein mediators of PCa pathogenesis, offering actionable insights into diagnostic and therapeutic innovation.

Microseminoprotein-beta (MSMB), the second most abundant prostate-secreted protein after PSA, (Lilja & Abrahamsson, 1988 [\[15\]](#)) has been shown to control prostate cell growth by regulating apoptosis (Garde et al., 1999 [\[16\]](#)). Unlike PSA, MSMB expression is not directly regulated by androgens. This androgen independence makes it possible for MSMB to be a supplementary biomarker for PSA, particularly in contexts where androgen receptor signaling is dysregulated, such as castration-resistant prostate cancer (CRPC) (Sjöblom et al., 2016 [\[17\]](#)). Our study found that MSMB demonstrated the strongest protective association, and this finding was validated in prospective studies (Byrne et al., 2019 [\[18\]](#); Haiman et al., 2013 [\[19\]](#)), supporting a potential protective role of MSMB in PCa. Some studies reported the level of MSMB in benign prostate tissues or BPH

was significantly elevated and decreased or lost in PCa (Luebke et al., 2019; Nam et al., 2006; Whitaker, Warren, Eeles, Kote-Jarai, & Neal, 2010), however, others paradoxically observed that the decreased expression of MSMB was both in the tumor (especially in more advanced tumor) and adjacent benign prostate tissue (Bergström, Järemo, Nilsson, Adamo, & Bergh, 2018). This apparent discrepancy may reflect dynamic compensatory regulation during disease progression. In early carcinogenesis, benign epithelia adjacent to low-grade tumors may upregulate MSMB as a protective response to counteract oncogenic stress, whereas advanced tumors drive epigenetic silencing of MSMB, propagating a permissive microenvironment for invasion. Overall, these findings position MSMB as a promising biomarker, which is expected to enhance the diagnostic and prognostic specificity of PCa through integration with PSA.

SERPINA3 is an inhibitor of serine proteases that was previously found to be upregulated in various types of cancer and its elevation was associated with a worse prognosis (De Mezer et al., 2023). Mechanistically, SERPINA3 is regulated by inflammatory cytokines so that its expression is increased in the inflammatory response (Péré-Brissaud et al., 2015), may drive tumor progression. SERPINA3 emerged as a significant risk factor for PCa in our MR analysis, which aligns with several proteomics (Nguyen et al., 2018; Zhang et al., 2022). The overexpression of SERPINA3 indicates that damage tends to occur in the body contributing to decreased cell adhesion ability and inhibition of apoptosis (Chelbi et al., 2012). These alterations may synergistically enhance metastatic potential, as evidenced by a recent study linking SERPINA3 to bone metastasis. A study identified that the enhanced expression of SERPINA3 stimulated the bone environment by promoting osteoblasts and osteoclasts activation, suggesting it may serve a diagnostic biomarker for PCa with bone metastasis phenotype and survival (Ito et al., 2023).

PRSS3, also named mesotrypsin, is reported to be aberrantly expressed in various types of tumors and participates in the progression and development of cancers. For example, PRSS3 was identified to be downregulated in lung cancer (Zhou, Li, & Luo, 2023) but upregulated in pancreatic cancer (Jiang et al., 2010; Xing et al., 2019), suggesting it may have different roles depending on the cellular or disease ways. A study demonstrated that PRSS3 was ectopically expressed in metastatic tissues of PCa and identified PRSS3 as a promising therapeutic target for metastatic PCa (Hockla et al., 2012). The research of PRSS3 in PCa is relatively limited, further studies are necessary to fully understand the functions and mechanisms of PRSS3 in human PCa. KLK3, which codes for PSA, is a well-known PCa biomarker. While its diagnostic utility is clinically validated, emerging evidences reveal that KLK3 mRNA detection in whole blood has a predictive role in tumor progression and therapeutic resistance, which is expected to become a potential prognostic marker for PCa (Boerrigter et al., 2021; Cho et al., 2024). KLK3's canonical role as a serine protease involves cleaving extracellular matrix (ECM) components and activating matrix metalloproteinases to promote tumor invasion and metastasis (Escaff et al., 2010; Moradi, Srinivasan, Clements, & Batra, 2019). Additionally, the androgen regulation of KLK3 expression correlated with the androgen receptor signaling pathway, a critical driver of PCa progression, particularly in CRPC (Lilja, Ulmert, & Vickers, 2008). Despite these insights, the precise biological pathways through which KLK3 contributes to PCa progression remain incompletely understood. Further research is needed to illustrate the direct and indirect effects of KLK3 on PCa cells and TME.

Despite substantial preclinical efforts identifying numerous promising biomarkers, their translation into routine clinical application remains strikingly limited due to the lack of further validation. Therefore, prospective validation is important and crucial.

Biological Mechanisms and Druggable Targets

Our integrative analysis revealed that the 193 candidate genes identified through MR were significantly enriched in inflammatory/immune pathways (e.g., cytokine-cytokine receptor interaction, JAK-STAT signaling) and cancer-related processes such as apoptosis and cell adhesion.

These pathways not only depict the molecular landscape of PCa but also identify viable targets for therapeutic intervention.

For instance, RRM2B is a DNA repair enzyme targeted by the clinical-stage drug TRIAPINE, and its role in maintaining genomic stability under replicative stress is consistent with the enrichment of the p53 signaling pathway.(Elfar et al., 2024 [DOI](#)) Similarly, the cell adhesion pathways is central to metastatic dissemination. As a member of Thy-1/ Ly-6 family involved in cell adhesion and metastasis, PSCA' s overexpression in high-grade PCa and metastatic lesions underscores the therapeutic potential of PSCA inhibitors in targeting tumor-stroma crosstalk to prevent metastatic spread.(Dizaj et al., 2024 [DOI](#)) Furthermore, HSPB1, a molecular chaperone implicated in apoptosis evasion and chemotherapy resistance(Garrido et al., 2006 [DOI](#); Hadaschik et al., 2008 [DOI](#); Kamada et al., 2007 [DOI](#)), aligns with the observed enrichment of survival pathways in aggressive PCa.(Shiota et al., 2013 [DOI](#); Vasiljevic et al., 2013 [DOI](#)) The antisense oligonucleotide APATORSEN, which silences HSPB1, restores chemosensitivity by reactivating apoptotic signaling, offering a strategy for therapeutic resistance against CRPC.(Le et al., 2023 [DOI](#)) Beyond these prioritized targets, the druggability analysis uncovered broader therapeutic potential. Proteins like RET (enriched in JAK-STAT signaling) and FGFR3 (linked to pathways in cancer) are already targeted by approved therapies in other malignancies, suggesting rapid repurposing potential for PCa. In conclusion, bridging pathway enrichment with druggability data transforms genetic associations into therapeutic hypotheses.

Study Strengths and Limitations

This study possessed several strengths, including the use of the largest collection of plasma protein data (covering more than 4800 proteins), large sample sizes of GWAS studies, a mutual validation across two independent outcome datasets and the use of colocalization analysis to support the MR results. Additionally, our assessment of the human blood proteome relied on two technologies (SOMAmer and Olink), which were valuable for identifying plasma proteins associated with diseases traits such as PCa.

Some limitations of our analysis should be acknowledged. Firstly, this investigation was only confined to Europeans, restricting the applicability of our findings to other populations. It is crucial for future studies to identify risk proteins in more diverse populations, especially African ancestry who face a higher risk of PCa. Secondly, our study concentrated on the proteins available in the MR analysis, which likely led to the omission of other potential therapeutic targets. Thirdly, we utilized proteomic data from Icelanders whose genetic backgrounds may differ from other European populations, which might introduce bias. Lastly, our analysis relied exclusively on publicly available GWAS summary statistics from openGWAS and FinnGen, which did not provide individual-level data on covariates, resulting in no direct assessment of demographic or clinical differences between cases and controls. However, this potential bias might be minimal, as we found that 20 proteins showing significance ($P < 0.05$) in both the deCODE and UKB-PPP studies had consistent associations with PCa.

In conclusion, we conducted a large-scale PW-MR study using the MR and colocalization analysis to investigate the genetic associations of up to 3,722 unique proteins with PCa. We revealed the complex genetic architecture of PCa and identified many new plasma proteins with strong causal associations to PCa. These findings highlighted the potential biomarkers for early detection and therapeutic targets, providing a foundation for future research and potential clinical applications.

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Additional information

Data availability

The GWAS summary statistics of blood pQTLs from deCODE can be found at <https://www.decode.com/summarydata/>, and those of the UKB-PPP can be found at <https://www.synapse.org/#!Synapse:syn51365303>. Summary statistics of PCa from the PRACTICAL consortium can be obtained from OpenGWAS (<https://gwas.mrcieu.ac.uk/>), and those of the FinnGen study can be obtained from <https://r10.finnngen.fi/>. The OpenTargets data were downloaded directly from the OpenTargets website (<https://platform.opentargets.org/downloads/data>).

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Additional files

Supplementary figures [↗](#)

Supplementary tables [↗](#)

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Reviewer #1 (Public review):

Summary:

In Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization the authors present a manuscript which seeks to identify novel markers for prostate cancer through analysis of large biobank-based datasets, and to extend this analysis to potential therapeutic targets for drugs. This is an area which is already extensively researched, but remains important, due to the high burden and mortality of prostate cancer globally.

Strengths:

The main strengths of the manuscript are the identification and use of large biobank data assets, which provide large numbers of cases and controls, essential for achieving statistical power. The databases used (deCODE, FinnGen and the UK Biobank) allow for robust numbers of cases and controls. The analytical method chosen, Mendelian Randomization, however, may not be appropriate to the problem (without extensive validation, MR can be prone to false or misleading discoveries). The manuscript also integrates multi-omic datasets, here using protein data as well as GWAS sources to integrate genomic and proteomic data.

Weaknesses:

The main weaknesses of the manuscript relate to the following areas:

(1) The failure of the study to analyse the data in the context of other closely related conditions such as benign prostatic hyperplasia (BPH) or lower urinary tract symptoms (LUTS), which have some pathways and biomarkers in common, such as inflammatory pathways (including complement) and specific markers such as KLK3. As a consequence, it is not possible for readers to know whether the findings are specific to prostate cancer, or whether they are generic to prostate dysfunction. Given the prevalence of prostate dysfunction (half of men reaching their sixth decade), the potential for false positives and overtreatment from non-specific biomarkers is a major problem, resulting in the evidence presented in this manuscript being weak. Other researchers have addressed this issue using the same data sources as presented here, for example in this paper looking at BPH in the UK Biobank population.

<https://www.nature.com/articles/s41467-018-06920-9>

(2) There is no discussion of Gleason scores with regard to either biomarkers or therapies, and a general lack of discussion around indolent disease as compared with more aggressive variants. These are crucial issues with regard to the triage and identification of genomically aggressive localized prostate cancers. See for example the work set out in: <https://doi.org/10.1038/nature20788>. In the revised version of the manuscript the authors set this out as a limitation, but this does not solve the core problem, which is that without this important biological context, the findings are unlikely to be robust.

(3) An additional issue is that the field of PCa research is fast-moving. The manuscript cites ~80 references, but too few of these are from recent studies and many important and relevant papers are not included. The manuscript would be much stronger if it compared and contrasted its findings with more recent studies of PCa biomarkers and targets, especially those concerned with multi-omics and those including BPH. In the latest revised version of the manuscript, some changes have been made, but the source data are still too limited for in-depth analysis.

(4) The Methods section provides no information on how the Controls were selected. There is no Table providing cohort data to allow the reader to know whether there were differences in age, BMI, ethnic grouping, social status or deprivation, or smoking status, between the Cases and Controls. These types of data are generally recorded in Biobank data; in the latest version of the manuscript the authors state that they don't have any ability to derive matched data, which again prevents deep analysis of the data.

Assessing impact:

Because of the weaknesses of the approach identified above, without further additions to the manuscript, the likely impact of the work on the field is minimal. There is no significant utility of the methods and data to the community, because the data are pre-existing and are not newly introduced to the community in this work, and mendelian randomization is a well-described approach in common use, and therefore the assets and methods described in the manuscript are not novel. In addition, Mendelian randomization is not always appropriate, especially when analysing publicly available data, see:

Stender et al. *Lipids in Health and Disease* (2024) 23:286

<https://doi.org/10.1186/s12944-024-02284-w>

With regard to the authors achieving their aims, without assessing specificity and without setting their findings in the context of the latest literature, the authors (and readers) cannot know or assess whether the biomarkers identified or the druggable targets will be useful in the clinic.

In conclusion, adding additional context and analysis to the manuscript would both help readers interpret and understand the work, and would also greatly enhance its significance. For example, the UK Biobank includes data on men with BPH / LUTS, as analysed in this paper, for example, <https://doi.org/10.1038/s41467-018-06920-9>. In the latest version of the manuscript and through the responses to earlier review comments, the authors explain why this has not been possible, but this naturally limits the value of the research.

<https://doi.org/10.7554/eLife.101584.2.sa3>

Reviewer #2 (Public review):

This is potentially interesting work, but the analyses are attempted in a rather scattergun way, with little evident critical thought. The structure of the work (Results before Methods) can work in some manuscripts, but it is not ideal here. The authors discuss results before we know anything about the underlying data that the results come from. It gives the impression that the authors regard data as a resource to be exploited, without really caring where the data comes from. The methods can provide meaningful insights if correctly used, but while I don't have reasons to doubt that the analyses were conducted correctly, findings are presented with little discussion or interpretation. No follow-up analyses are performed.

This is much improved but there remain some small concerns and one large concern:

Using numbering from the previous review:

(1) This looks better, but I still don't understand the claim in the text: "We found 5 genetic risk loci contained at least one SNP passing the genome-wide significance threshold of $P \leq 5 \times 10^{-8}$ ". Far more gene regions appear to cross 10^{-8} in Figure 1. What am I missing?

(6) I don't understand the authors' response here. Early detection is important, but MR is not the right tool to investigate biomarkers for early detection. Biomarkers for early detection do not have to be causal biomarkers. The authors replied to this point, but the manuscript was unchanged.

(7) Again, the authors still state "193 proteins were associated with PCa risk" even though they acknowledge that their analysis does not test whether proteins associate with PCa risk or not. When an error is pointed out, and you acknowledge it, please change the manuscript to correct the text. Otherwise, what is the peer review process for?

The large concern is that these analyses, while now better explained, are still the product of a semi-automated procedure. It is a good procedure, but the manuscript essentially takes public data from different sources and uses this to create a manuscript. Overall, I think there is enough novel synthesis to justify publication, but it is not automatic.

Strengths:

The data and methods used are state-of-the-art.

Weaknesses:

The reader will have to provide their own translational insight.

<https://doi.org/10.7554/eLife.101584.2.sa2>

Reviewer #3 (Public review):

Summary of concerns about the revised submission from the Reviewing Editor:

With respect to Originality of the work, in the last 18 months, there have been 38 publications on combined topics of: (i) UK Biobank data, (ii) Mendelian randomization, (iii) and prostate cancer. The authors should consider the literature addressing prostate cancer via Mendelian randomization--specifically those using the UK Biobank data--published from 2024 onwards. A proper and comprehensive synthesis of recent findings should be made, to allow readers to compare and contrast how the work supports (or differs) from the findings presented in these other published studies.

With respect to the significance of the findings, we feel the study data are incomplete for the strength of evidence. Given the deluge of manuscripts and publications on similar topics, the study offers incremental evidence and lacks a synthesis of all currently published findings.

<https://doi.org/10.7554/eLife.101584.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In Causal associations between plasma proteins and prostate cancer: a Proteome-Wide Mendelian Randomization, the authors present a manuscript which seeks to identify novel markers for prostate cancer through analysis of large biobank-based datasets and to extend this analysis to potential therapeutic targets for drugs. This is an area that is already extensively researched, but remains important, due to the high burden and mortality of prostate cancer globally.

Strengths:

The main strengths of the manuscript are the identification and use of large biobank data assets, which provide large numbers of cases and controls, essential for achieving statistical power. The databases used (deCODE, FinnGen, and the UK Biobank) allow for robust numbers of cases and controls. The analytical method chosen, Mendelian Randomization, is appropriate to the problem. Another strength is the integration of

multi-omic datasets, here using protein data as well as GWAS sources to integrate genomic and proteomic data.

Thank you for your positive feedback regarding the overall quality of our work and we greatly appreciate you taking time and making effort in reviewing our manuscript.

Weaknesses:

The main weaknesses of the manuscript relate to the following areas:

(1) The failure of the study to analyse the data in the context of other closely related conditions such as benign prostatic hyperplasia (BPH) or lower urinary tract symptoms (LUTS), which have some pathways and biomarkers in common, such as inflammatory pathways (including complement) and specific markers such as KLK3. As a consequence, it is not possible for readers to know whether the findings are specific to prostate cancer or whether they are generic to prostate dysfunction. Given the prevalence of prostate dysfunction (half of men reaching their sixth decade), the potential for false positives and overtreatment from non-specific biomarkers is a major problem, resulting in the evidence presented in this manuscript being weak. Other researchers have addressed this issue using the same data sources as presented here, for example, in this paper, looking at BPH in the UK Biobank population. <https://www.nature.com/articles/s41467-018-06920-9>

Thank you for your valuable comment. We fully agree that biomarker development must prioritize specificity to avoid overtreatment. While our study is a foundational step toward identifying potential therapeutic targets or complementary biomarkers for prostate cancer—not as a direct endorsement of these proteins for standalone clinical diagnosis. Mendelian randomization analysis strengthens causal inference by design, and we further ensured robustness through sensitivity analyses (e.g., MR-Egger regression for pleiotropy, Bonferroni correction for multiple testing). These methods distinguish true causal effects from nonspecific associations. Importantly, while PSA's lack of specificity is widely recognized, its role in reducing PCa mortality underscores the value of biomarker-driven screening. Our findings align with the need to integrate multiple markers (e.g. combining a novel protein with PSA) to improve diagnostic precision. Translating these causal insights into clinical tools remains challenging but represents a necessary next step, and we emphasize that this work provides a rigorous starting point for future validation studies.

(2) There is no discussion of Gleason scores with regard to either biomarkers or therapies, and a general lack of discussion around indolent disease as compared with more aggressive variants. These are crucial issues with regard to the triage and identification of genomically aggressive localized prostate cancers. See, for example, the work set out in: <https://doi.org/10.1038/nature20788>

Thank you for pointing this out. We acknowledge that our original analysis did not directly address this critical issue due to a key data limitation: the publicly available GWAS summary statistics for PCa (from openGWAS and FinnGen) do not provide genetic associations stratified by phenotypic severity or molecular subtypes. This limitation precluded MR analysis of proteins specifically linked to aggressive disease. To partially bridge this gap, we integrate evidence from recent studies in the revised Discussion section to explore the relevance of potential biomarkers to aggressive PCa.

(3) An additional issue is that the field of PCa research is fast-moving. The manuscript cites ~80 references, but too few of these are from recent studies, and many important and relevant papers are not included. The manuscript would be much stronger if it

compared and contrasted its findings with more recent studies of PCa biomarkers and targets, especially those concerned with multi-omics and those including BPH.

Thank you for your professional comments. We have rigorously updated the manuscript to include more recent publications and we systematically compare and contrast our findings with these recent studies in the revised Discussion section.

(4) The Methods section provides no information on how the Controls were selected. There is no Table providing cohort data to allow the reader to know whether there were differences in age, BMI, ethnic grouping, social status or deprivation, or smoking status, between the Cases and Controls. These types of data are generally recorded in Biobank data, so this sort of analysis should be possible, or if not, the authors' inability to construct an appropriately matched set of Controls should be discussed as a Limitation.

We thank the reviewer for raising this important methodological concern. We have expanded the Limitations section to state it.

“Lastly, our analysis relied exclusively on publicly available GWAS summary statistics from openGWAS and FinnGen, which did not provide individual-level data on covariates, resulting in no direct assessment of demographic or clinical differences between cases and controls.”

Reviewer #2 (Public review):

This is potentially interesting work, but the analyses are attempted in a rather scattergun way, with little evident critical thought. The structure of the work (Results before Methods) can work in some manuscripts, but it is not ideal here. The authors discuss results before we know anything about the underlying data that the results come from. It gives the impression that the authors regard data as a resource to be exploited, without really caring where the data comes from. The methods can provide meaningful insights if correctly used, but while I don't have reasons to doubt that the analyses were conducted correctly, findings are presented with little discussion or interpretation. No follow-up analyses are performed.

In summary, there are likely some gems here, but the whole manuscript is essentially the output from an analytic pipeline.

We thank the reviewer for the thoughtful evaluation of our work. In response to the concerns regarding manuscript structure and interpretative depth, we have restructured the manuscript to present the Methods section before Results, ensuring transparency in data sources and analytical workflows. Additionally, the Discussion section has been substantially revised to provide mechanistic explanations for key findings (e.g., associated phenotype, causal proteins, druggable targets), contextualize results within recent multi-omics studies and highlight clinical implications. These revisions aim to transform the work from a pipeline-driven analysis to a biologically grounded investigation, offering actionable insights into prostate cancer pathogenesis and therapeutic development.

Taking the researchers aims in turn:

(1) Meta-GWAS - while combining two datasets together can provide additional insights, the contribution of this analysis above existing GWAS is not clear. The PRACTICAL consortium has already reported the GWAS of 70% of these data. What additional value does this analysis provide? (Likely some, but it's not clear from the text.) Also, the presentation of results is unclear - authors state that only 5 gene regions contained variants at $p < 5 \times 10^{-8}$, but Figure 1 shows dozens of hits above 5×10^{-8} . Also, the red line in Figure 1 (supposedly at 5×10^{-8}) is misplaced.

Thank you very much for your feedback. Although the PRACTICAL consortium constituted the majority of PCa GWAS data, our meta-analysis integrating FinnGen data enhanced statistical power enabling robust detection of low-frequency variants with minor allele frequencies. Moreover, FinnGen's Finnish ancestry (genetic isolate) helps distinguish population-specific effects. The presentation of results showed the top 5 gene regions contained variants at $p < 5 \times 10^{-8}$. We apologize for not noticing that the red line was not displayed correctly in the original figures included in the manuscript. We have updated it in the revised manuscript.

(2) Cross-phenotype analysis. It is not really clear what this analysis is, or why it is done. What is the iCPAGdb? A database? A statistical method? Why would we want to know cross-phenotype associations? What even are these? It seems that the authors have taken data from an online resource and have written a paragraph based on this existing data with little added value.

We appreciate the opportunity to clarify this analysis. The cross-phenotype analysis was designed to systematically identify phenotypic traits that share genetic or molecular pathways with prostate cancer, thereby uncovering pleiotropic mechanisms or shared risk factors. Here, iCPAGdb (integrated Cross-Phenotype Association Genetics Database) is a curated repository that aggregates GWAS summary statistics and evaluates genetic pleiotropy using LD-proxy associations from the NHGRI-EBI GWAS Catalog. Prostate carcinogenesis involves multisystem interactions, including spanning endocrine dysregulation, immune microenvironment remodeling and metabolic reprogramming, rather than isolated molecular pathway disruptions. Therefore, it is indispensable for discriminating primary pathogenic drivers from secondary compensatory responses, ultimately informing the development of precision therapeutic strategies.

In response to your concerns, we have revised the Results section to explicitly define the rationale and methodology of cross-phenotype analysis and restructured the Discussion to interpret phenotype-PCa associations within unified biological frameworks (e.g., metabolic dysregulation, androgen signaling), rather than presenting them as isolated findings.

(3) PW-MR. I can see the value of this work, but many details are unclear. Was this a two-sample MR using PRACTICAL + FinnGen data for the outcome? How many variants were used in key analyses? Again, the description of results is sparse and gives little added value.

We thank you for raising this issue. Two-sample MR refers to an analytical design where genetic instruments for the exposure (plasma proteins) and genetic associations with the outcome (PCa) are derived from non-overlapping populations. This ensures complete sample independence between exposure and outcome datasets to avoid confounding biases, regardless of whether the outcome data originate from single or multiple cohorts. The meta-analysis of PRACTICAL and FinnGen GWAS generates 27,210 quality-controlled variants ($p < 5 \times 10^{-8}$, $MAF \geq 1\%$, LD-clumped $r^2 < 0.1$) used in key analyses. Regarding the concern about sparse interpretation, we have substantially expanded the Discussion by comparing significant protein findings (e.g., MSMB, SERPINA3) with results from existing functional studies and multi-omics datasets and unravelling new insights.

(4) Colocalization - seems clear to me.

(5) Additional post-GWAS analyses (pathway + druggability) - again, the analyses seem to be performed appropriately, although little additional insight other than the reporting of output from the methods.

The post-MR druggability and pathway analyses serve two primary scientific purposes: (1) therapeutic prioritization - systematically evaluating which MR-identified proteins represent tractable drug targets (either through existing FDA-approved agents or compounds in clinical development) with direct relevance to cancer or PCa management, and (2) mechanistic hypothesis generation - mapping these candidate proteins to coherent biological pathways to guide future functional validation studies investigating their causal roles in prostate carcinogenesis. In response to your feedback, we have restructured the Discussion section under the subheading “Biological Mechanisms and Druggable Targets” to synthesize these findings, explicitly linking biological pathway to therapeutic targets.

Minor points:

(6) The stated motivation for this work is "early detection". But causality isn't necessary for early detection. If the authors are interested in early detection, other analysis approaches are more appropriate.

We appreciate your insightful feedback. Early detection is one motivation for this work, meanwhile, our goal is also to identify causally implicated proteins that may serve as intervention targets for PCa prevention or therapy. Establishing causality is critical for distinguishing biomarkers that drive disease pathogenesis from those that are secondary to disease progression, as the former holds greater specificity for early detection and prioritization of therapeutic targets. While we acknowledge that validation for early detection may require additional methodologies, MR analysis provides a foundational step by prioritizing candidate proteins with causal links to disease. This approach ensures that downstream efforts focus on biomarkers and targets with the greatest potential to alter disease trajectories, rather than merely correlative markers.

(7) The authors state "193 proteins were associated with PCa risk", but they are looking at MR results - these analyses test for disease associations of genetically-predicted levels of proteins, not proteins themselves.

True, in MR, the exposure of interest is the lifelong effect of genetically predicted protein levels. This approach is designed to infer causality while avoiding confounding and reverse causation, as genetic variants are fixed at conception and unaffected by disease processes. When we state “193 proteins were associated with PCa risk,” we specifically refer to proteins whose genetically predicted levels (based on instrument SNPs from protein QTLs) show causal links to PCa. Importantly, MR does not measure the direct association between observed protein concentrations and disease. Instead, it estimates the lifelong causal effect of protein levels predicted by genetics. This distinction is critical for disentangling cause from consequence. For example, a protein elevated due to tumor progression would not be identified as causal in MR if its genetic predictors are unrelated to PCa risk.

We acknowledge that clinical translation requires further validation of these proteins in observational studies measuring actual protein levels. However, MR provides a robust first step by prioritizing candidates with causal roles, thereby reducing the risk of investing in biomarkers confounded by disease processes.

Reviewer #1 (Recommendations for the authors):

As outlined above, the major weakness of the manuscript is its failure to consider BPH / LUTS, and whether the markers and targets are specific to PCa or not. Specific improvements that the authors could consider might include a literature review of the features identified for their 20 high-risk proteins, and ideally also analyze whether these proteins are upregulated or downregulated in the databases they have analysed (for

example it will be easy to analyze whether these proteins are dysregulated in BPH patients as these are specifically identified in the UK Biobank).

The authors may be able to gain context for this approach by looking at papers analyzing BPH and the complement cascade and other proteins from the authors' top 10 or top 20, for example: <https://doi.org/10.1002/pros.24639>IF: 2.6 Q2

Other sources can be identified by examining the literature for recent omics papers analysing BPH, especially those that analyse and compare BPH / PCa specifically.

Thank you for highlighting the critical need to distinguish PCa-specific biomarkers from those shared with BPH. In response, we conducted a literature review of multi-omics datasets and prospective cohort studies, systematically evaluating the specificity of prioritized proteins by comparing their expression trends in PCa and BPH or benign prostate tissues. These findings are now integrated into the revised Discussion section under the subheading "Plasma Proteins Causal Links to Prostate Cancer".

In the Discussion, the paragraph (line 288) on PSA is extremely weak. The authors state that further research is needed, and yet only reference four articles (from 2008, 2010, 2012, 2014), none of which are from the last decade. Considerable amounts of research from the last ten years have been published on PSA, for example, see this article from 2018, which specifically analyses PSA in the context of the UK Biobank. This section should be made more up-to-date with the latest literature findings. <https://doi.org/10.1038/s41467-018-06920-9>

Thank you very much for your feedback. We acknowledge the need to strengthen the discussion on PSA by incorporating recent literature. In the revised manuscript, we have expanded the PSA discussion to integrate contemporary research on the prognostic role of PSA in the progression of PCa and its limitations in cancer screening, ensuring that our discussion reflected the current consensus and controversies.

Also in the Discussion, the analysis of phenotypic indicators is insufficiently comprehensive and should reference other recent research. For example, this recent UK Biobank study dealt with a wide range of conditions, including prostate cancer, and identified similar factors to those identified in this paper. The authors should compare and contrast their phenotypic findings with the existing literature. <https://doi.org/10.1038/s41588-024-01898-1>

Thank you for addressing the comprehensiveness of phenotypic analysis. We have learned recent large-scale phenome-wide analyses (linked in your feedback) that explore multi-omics biomarkers and their associations with a range of different diseases. We have compared and contrasted our phenotypic findings with the existing literature and revised the Discussion section to interpret phenotype-PCa associations, emphasizing both shared pathways and disease-specific signals.

Under Methods, there is too little information on how Controls were selected, whether any matching process was conducted, or whether there are fundamental differences between the cases and controls (such as smoking status, BMI, comorbidities). The authors use R, and a library such as MatchIt could be used to ensure that the Controls cohort is appropriately matched to the Cases.

As outlined above, we acknowledge that our original analysis did not directly address this critical issue due to a key data limitation. The publicly available GWAS summary statistics for PCa (from openGWAS and FinnGen) do not provide individual-level data on covariates,

resulting in no direct assessment of demographic or clinical differences between cases and controls.

An important final point is that, as far as I can tell, no UK Biobank Application Number has been specified in the manuscript. This is vital to establish that there was an original hypothesis being investigated (as opposed to data dredging of open access resources), especially in light of the largely mechanistic flow of the manuscript and lack of PCa and relevant confounder-specific discussion. The authors may be aware of the work of Stender et al (2024) regarding formulaic papers using Mendelian randomization, especially that "[All] combinations of exposure and outcome results based on data available in IEU openGWAS (<https://gwas.mrcieu.ac.uk/>) can be browsed online on epigraphDB.org. In other words, these results are, in effect, already published. Reporting them again in a scientific paper adds nothing to what can be looked up online in minutes." The authors may wish to address this issue directly.

*Stender, S., Gellert-Kristensen, H. & Smith, G.D. Reclaiming Mendelian randomization from the deluge of papers and misleading findings. *Lipids Health Dis* 23, 286 (2024). <https://doi.org/10.1186/s12944-024-02284-w>*

We confirm that all data used in this study were obtained from publicly available GWAS summary statistics (e.g., PRACTICAL consortium, FinnGen) and proteomic datasets (deCODE, UKB-PPP). Our research was guided by a predefined hypothesis to investigate causal plasma protein biomarkers for prostate cancer, rather than exploratory data mining. The analytical pipelines and integrative approaches (e.g., colocalization, druggability assessment) were specifically designed to address this hypothesis, aligning with the ethical use of open-access resources.

Reviewer #2 (Recommendations for the authors):

There are several specific recommendations in the public review (e.g., clarify the contribution of the GWAS). Otherwise, there is nothing clearly incorrect, but translational insight is missing - the analyses are not clearly connected to the scientific literature. This is a limitation rather than a flaw - the manuscript will likely still be useful to readers.

We thank you for highlighting the need to strengthen translational insights and contextualize our findings within existing literature. In the revised manuscript, we have expanded the Discussion section to systematically compare our results with prior mechanistic and clinical studies, including the shared pathways of associated phenotypes, the potential of significant proteins in biomarkers and therapeutic targeting. These revisions ensure our analyses are firmly rooted in the scientific literature.

<https://doi.org/10.7554/eLife.101584.2.sa0>